



## David Munoz Tord

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## Core Skills

|            |          |
|------------|----------|
| R          | 10 years |
| SQL        | 5 years  |
| Python     | 5 years  |
| Bash       | 7 years  |
| JavaScript | 5 years  |

## Stack

- ▶ Shiny (golem, rhino, leprechaun)
- ▶ devtools, testthat, roxygen2, renv
- ▶ tidyverse, data.table, ggplot2
- ▶ Scikit-learn, PyTorch
- ▶ Docker, GitHub Actions, CircleCI

## Platforms

- ▶ Linux administration (Ubuntu, RHEL-like)
- ▶ Cloud infrastructure (AWS)
- ▶ Posit Connect / Posit Workbench deployments
- ▶ CI/CD and container workflows
- ▶ PostgreSQL, Jira, Git

## Focus Areas

- ▶ Data platform deployment and support
- ▶ Data engineering and analytics delivery
- ▶ Linux systems and secure environments
- ▶ Troubleshooting and incident response

## Profile

Infrastructure and analytics engineer with 5+ years of experience delivering data science platforms in regulated and client-facing environments. Strong background in Linux, cloud deployment (AWS), CI/CD, containerized services, and production support for R/Python analytics workflows. I combine hands-on technical delivery with clear stakeholder communication, end-to-end ownership, and a consulting mindset.

## Experience

### Senior R Developer

02/2025 - Present

#### Johnson & Johnson

- Co-led infrastructure and platform migration from SAS-based workflows to a modern R stack for clinical reporting teams
- Built and maintained production R packages and release pipelines
- Designed and operated containerized Posit and Shiny infrastructure for secure, reproducible execution across development and production environments
- Implemented an MCP server and internal code-context integrations so LLM agents could understand the team stack and assist clinical programmers with output generation
- Owned DevOps delivery for the R team: CI/CD maintenance, deployment automation, operational troubleshooting, and platform reliability tasks

### Data Science Consultant

01/2024 - 02/2025

#### Freelance - Pharma & Tech

- Delivered end-to-end client projects from infrastructure scoping and architecture design to deployment and support handover
- Deployed custom R Shiny and API solutions on AWS with CI/CD pipelines, containerization, and operational documentation
- Automated data pipelines and reporting workflows with Bash/Python scripting and reproducible environments
- Managed concurrent client requests and communicated technical trade-offs and maintenance plans to non-technical audiences

### Full-Stack Data Scientist

04/2022 - 01/2024

#### FlowBank

- Built a custom data science and machine learning platform from scratch on AWS for analytics and DS teams
- Set up data scientist IDE environments (notebook-based workflows) with standardized project templates and access controls
- Containerized notebooks and analytics services with Docker to guarantee reproducibility across development, validation, and production environments
- Implemented CRON-scheduled jobs for model retraining, batch scoring, report generation, and maintenance tasks
- Maintained machine learning pipelines in production
- Delivered internal tooling and Linux-based automation (R, Bash, Python), with operational documentation and onboarding support for business teams

### Biostatistics Instructor

09/2020 - 01/2022

#### University of Geneva

- Taught statistics and practical R/Python workshops to mixed technical audiences
- Produced structured technical learning material and clear troubleshooting guides

### Neuroscience Researcher

03/2020 - 01/2022

#### Campus Biotech

- Developed large-scale data processing pipelines for clinical and neuroimaging datasets
- Designed and operated an Apache Spark cluster on bare-metal Linux machines for distributed preprocessing and model experimentation workloads

- Technical communication with clients

## Languages

|          |         |
|----------|---------|
| French   | Native  |
| English  | Fluent  |
| Spanish  | Native  |
| Italian  | Passive |
| Japanese | Passive |

## Education

### PhD Candidate in Computational Neuroscience

03/2020 - 01/2022

EPFL - Switzerland

Topic: effects of sensory modulation in food behavior

### MSc in Computational Neuroscience

09/2018 - 03/2020

University of Geneva - Switzerland

Thesis: denoising high-resolution fMRI signals with tICA

### Additional Studies in Computer Science

09/2017 - 06/2018

UMass Amherst - USA

Project: machine learning for neuroimaging data

### BSc in Cognitive Science

09/2014 - 06/2017

University of Geneva - Switzerland

Thesis: early detection of autism-related event potentials

## Community

- 🏆 Top 12.5% on [GitHub](#) (2026)
- 🔗 Top 2% on [StackOverflow](#) (2024)
- 🌐 Global Fellow scholarship (2017)
- 🚲 Co-founder of [Rocket-Courrier](#), a sustainable bike courier cooperative.

## Source

- 🔗 [Quarto source code](#)

- Applied advanced statistical and machine learning methods in reproducible computing environments
- Collaborated with international teams and maintained strict data governance standards

## Open Source and Community

See my [GitHub profile](#) for a broader list of projects.

### Author and Lead Maintainer

R, Python, Shiny

- **juncO** (Johnson & Johnson): built a clinical reporting package for production-grade TLG generation and document outputs
- **circle (rOpenSci)**: official maintainer of the CircleCI integration package for R
- **echarts4r** and **fiRebase** (Opifex): maintainer; contributed to interactive visualization tooling and secure backend integration
- Created cross-functional tools including **DbVieweR** and **MathAnimatoR** for reusable analytics workflows
- Contributed to widely used ecosystem projects including **ggplot2**, **brms**, and **AFNI**

### Co-President

Training, Evangelism, Data Science

- Co-created **Typr**, a modern type-system initiative for improving long-term R package quality and maintainability
- Designed and delivered community workshops, with all material published as [open training resources](#)
- Organized recurring trainings, hackathons, and local R community sessions at the University of Geneva

## Selected Publications

See my [publication page](#) for a full list.

**Investigating the effect of liraglutide on self-reported liking and neural responses to food stimuli.** Coppin, G., Munoz Tord, D., Pool, Cereghetti, D., Golay, A., Sander, D., & Pataky, Z. *Journal of Obesity* (2023) - [Paper](#)

**Differential contributions of ventral striatum subregions to the motivational and hedonic components of the affective processing of reward.** Pool, E. R., Munoz Tord, D., Delplanque, S., Stussi, Y., Cereghetti, D., Vuilleumier, P., & Sander, D. *Journal of Neuroscience* (2022) - [Paper](#)

**3D-printed pacifier-shaped mouthpiece for fMRI-compatible gustometers.** Munoz Tord, D., Coppin, G., Pool, E. R., Mermoud, C., Pataky, Z., Sander, D., & Delplanque, S. *Eneuro* (2021) - [Paper](#)

**Early spatial attention deployment towards aggressive voices.** Burra, N., Kerzel, D., Munoz Tord, D., Grandjean, D., & Ceravolo, L. *Social Cognitive and Affective Neuroscience* (2019) - [Paper](#)

## Professional References

[Gabriel Becker](#) ✉, Statistical Computing Consultant and R-core contributor

[Joseph Kovach](#) ✉, Head of Technological Solutions at Johnson & Johnson

[John Coene](#) ✉, Lead Developer at Opifex

[Pierre Saouter](#) ✉, Head Data Scientist at FlowBank